CSE 400 — Fundamentals of Probability in Computing

Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

1. Marginal PMF

1.1 Definition

For discrete random variables (X) and (Y), the **joint PMF** is

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The **marginal PMFs** are obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

1.2 Marginal PMF from a Joint PMF Table

Row sums give $p_X(x)$, and column sums give $p_Y(y)$.

$$p_X(x_i) = \sum_j p_{ij}$$

$$p_Y(y_j) = \sum_i p_{ij}$$

Normalization conditions:

$$\sum_i p_X(x_i) = 1$$

$$\sum_j p_Y(y_j) = 1$$

1.3 Example

Let the joint PMF be

$p_{XY}(x,y)$	$y=0$	$y=1$
$x=0$	$1/8$	$1/4$
$x=1$	$3/8$	$1/4$

Step-by-step computation

Compute $p_X(0)$

$$p_X(0) = \frac{1}{8} + \frac{1}{4}$$

$$= \frac{1}{8} + \frac{2}{8}$$

$$= \frac{3}{8}$$

Compute $p_X(1)$

$$p_X(1) = \frac{3}{8} + \frac{1}{4}$$

$$= \frac{3}{8} + \frac{2}{8}$$

$$= \frac{5}{8}$$

Compute $p_Y(0)$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8}$$

$$= \frac{4}{8}$$

$$= \frac{1}{2}$$

Compute $p_Y(1)$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4}$$

$$= \frac{1}{2}$$

Final Marginal PMFs

$$p_X(x) = \begin{cases} 3/8 & x = 0 \\ 5/8 & x = 1 \end{cases}$$

$$p_Y(y) = \begin{cases} 1/2 & y = 0 \\ 1/2 & y = 1 \end{cases}$$

2. Correlation

2.1 Definition

A basic way to measure joint behavior is through the **product moment**.

In the continuous case,

$$R_{XY} = E[XY]$$

It tells us the **average value of the product** (XY).

This is often called a **raw correlation measure**.

2.2 Properties

$$R_{XY} = E[XY]$$

- Depends on the **scale of** (X) **and** (Y)
- Computed **about the origin**, not about the means

Therefore it mixes together

- mean values
- spread
- dependence

If

$$R_{XY} = 0$$

then the variables are **orthogonal about the origin**.

To remove the effect of the means, we **center the variables first**, which leads to **covariance**.

3. Covariance

3.1 Definition and Interpretation

Covariance is defined as

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Interpretation
First subtract the mean from each variable.
Then measure how the **centered quantities vary together**.
Covariance captures whether deviations from the means occur

- in the **same direction**
- or in **opposite directions**

Sign Interpretation

- $Cov(X, Y) > 0 \rightarrow$ Positive covariance
- $Cov(X, Y) < 0 \rightarrow$ Negative covariance
- $Cov(X, Y) = 0 \rightarrow$ Zero covariance

Covariance tells the **direction of joint linear variation** between (X) and (Y).
However, the **magnitude depends on units** and cannot measure the strength of the relationship.

3.2 Algebraic Form

Start with

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand the product:

$$= E[XY - X\mu_Y - \mu_X Y + \mu_X\mu_Y]$$

Apply linearity of expectation:

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Since

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Therefore

$$Cov(X,Y) = E[XY] - \mu_X \mu_Y$$

or

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

3.3 Properties of Covariance

1. Symmetry

$$Cov(X,Y) = Cov(Y,X)$$

2. Variance as a Special Case

$$Cov(X,X) = Var(X)$$

3. Linearity with Respect to Constants

$$Cov(aX+b, cY+d) = ac, Cov(X,Y)$$

4. Independence Implies Zero Covariance
If

$$X \perp Y$$

then

$$Cov(X,Y) = 0$$

However,
zero covariance means uncorrelated, but it does not imply independence in general.

3.4 Example 1

Given

$$Y = -2X + 3$$

Find

$$Cov(X,Y)$$

Step 1: Use covariance formula

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

Step 2: Substitute expression for (Y)

$$Y = -2X + 3$$

Step 3: Compute (XY)

$$XY = X(-2X + 3)$$

$$= -2X^2 + 3X$$

Step 4: Compute expectation of (Y)

$$E[Y] = E[-2X + 3]$$

$$= -2E[X] + 3$$

Step 5: Substitute into covariance formula

$$Cov(X, Y) = E[-2X^2 + 3X] - E[X](-2E[X] + 3)$$

Expanding:

$$= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X]$$

$$= -2E[X^2] + 2(E[X])^2$$

Step 6: Use variance identity

$$Var(X) = E[X^2] - (E[X])^2$$

Thus

$$Cov(X, Y) = -2Var(X)$$

3.5 Pearson's Correlation Coefficient

Pearson's correlation coefficient measures the **effect of change in one variable when the other variable changes**.

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

or

$$\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Properties:

- Measures **strength and direction of linear relationship**

- **Normalized version of covariance**
- **Unit-free**

Range:

$$-1 \le \rho_{XY} \le 1$$

Interpretation:

- $\rho_{XY} > 0 \to$ positive linear trend
- $\rho_{XY} < 0 \to$ negative linear trend
- $\rho_{XY} = 0 \to$ no linear correlation

4. Jointly Gaussian Random Variables

4.1 Definition

A pair $((X, Y))$ is **jointly Gaussian** if

- its **joint density has the bivariate Gaussian form**
- the **marginals are Gaussian**

4.2 Algebraic Form

The **joint Gaussian PDF** is

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

Where

- (μ_X, μ_Y) = means
- (σ_X, σ_Y) = standard deviations
- (ρ) = Pearson correlation coefficient

4.3 Interpretation

Example from the lecture:

Relationship between

- (X): Atmospheric CO_2 concentration (ppm)
- (Y): Global surface temperature anomaly (°C)

Because both variables vary due to many random environmental processes such as

- solar radiation
- ocean circulation
- emissions
- weather variability

their joint behavior can often be approximated by a **Joint Gaussian Distribution**.